CURRENT ELECTRICITY

SUB TOPICS

**

Represents Conducting Wire where Potential difference Same

Same Potential.
Put a Random Variable (A) For Simplification...

- Ohm's Law
- Conductor
- Electric current
  - (i) InStantaneous
  - (ii) Average current
- Current density (J)
- Parallel and Series connections.
- Kirchoff's Law
- EMF...
- Average relaxation time ($\bar{\tau}$)
- Mobility (μ)
- Efficiency Of Cell (η)
- Power Rating of Bulbs
- Equivalent Resistance by Method Of Symmetry
- Meter Bridge Experiment.
- WheatStone Bridge
- Electrical Power consumption.
- Max Power Transfer theorm
- N identical cells (i) Series
  (ii) Parallel.
- Potentiometer, Galvanometer
- Conversion Of Galvanometer into a VoltMeter in Range (0-V) volts
- Amount Of Resistance Given to Shunt ...

Concepts
And
KEY POINTS :

* $i = \frac{dq}{dt}$

* $q = \int i\,dt$

* $i_{avg} = \frac{\int_{t_1}^{t_2} i\,dt}{\int_{t_1}^{t_2} dt}$

* Ohm's Law : $\boxed{V = iR}$

Current density (J):

$$J = \frac{i}{A}\ \text{Amp/m}^2 = \sigma E$$

$$J = \frac{neAV_d}{A} = neV_d$$

$\boxed{i = neAV_d}$ ~ General Current formula

σ : Conductivity $= J/E$
ρ : Resistivity

$$\rho = \frac{1}{\sigma}$$

Mobility $(\mu) = \frac{V_d}{E}$

⇨ $\boxed{\sigma = ne\mu}$

⇨ $V_d = \left(\frac{eE}{m}\right)\tau$ ($\because$ τ is Average relaxation time)

⇨ $\sigma = \frac{ne^2\tau}{m}$

Resistance:

$$R = \frac{\rho l}{A}$$

- Specific resistance is Same for two diff Body's of Same Material ...
- V_d is independent on diameter ...

* Parallel :

(i) → Diff
(V) → Same

* Series :

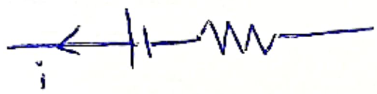
(i) → Same
(V) → Different ... //

☐ Electrical Power consumption :

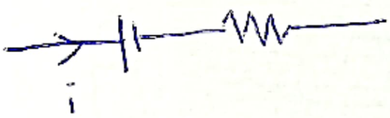
$P \times t$ = Heat transfer

$$\boxed{P \times t = ms\Delta T}$$

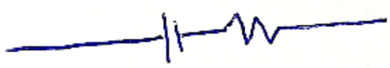
☐ Terminal P.D



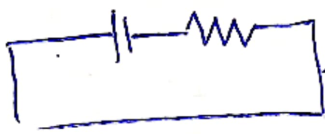
i. $\boxed{V_A - V_B = E - ir}$



ii. $\boxed{V_A - V_B = E + ir}$



iii. $\boxed{V_A - V_B = E}$ $(\because i = 0)$.



→ Pure conducting wire.

iv. $\boxed{V_A - V_B = 0}$

1. $\boxed{S = \frac{i_g G}{I - I_g}}$

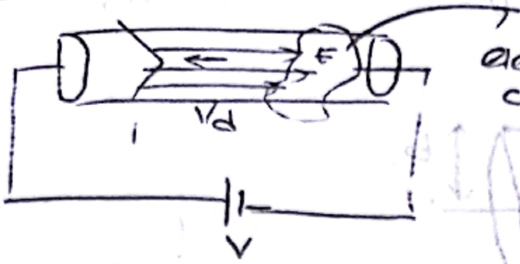
2. $\boxed{i_g(G+S) = V}$

Converting Moving Coil Gallvanometer into an

Ammeter Of Range (0-I) Amp. : Low in Parallel

Volt Meter Of Range (0-V) Volt : High in Series.

CURRENT ELECTRICITY ...



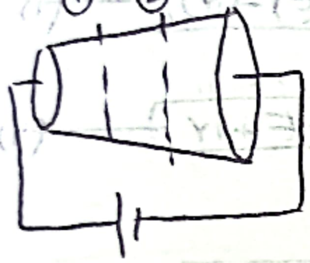
i. $i = neAV_d$

ii. $J = \frac{i}{A} = nev_d$.

iii. $\bar{J} = \sigma\bar{E}$ ($\because \sigma$ = conductivity)

iv. $\sigma = \frac{1}{\rho}$ Resistivity...

10.



i. $i_1 = i_2$

ii. $J_1 > J_2$

iii. $V_{d_1} > V_{d_2}$

iv. $E_1 > E_2$.

Mobility (μ)

$\mu = \frac{V_d}{E}$

$\sigma = \mu n e$.

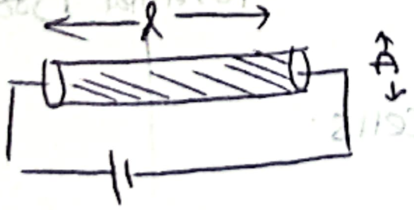
Mobility | Charge of each Electron ... | No. of e^-

$$\boxed{\sigma = \frac{ne^2\tau}{m}}$$

→ Relaxation time.

Time interval B/w Collisions to Collisions.

Uniform Conductor:



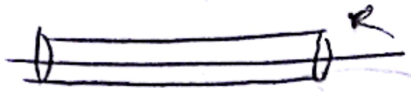
$J = \sigma E$

$\frac{i}{A} = \frac{1}{\rho}\frac{V}{l}$.

$\left(\frac{\rho l}{A}\right) i = V$

$\boxed{V = iR}$ $\boxed{R = \frac{\rho l}{A}}$

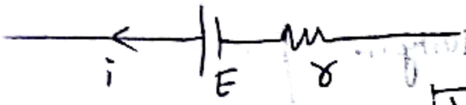
1.



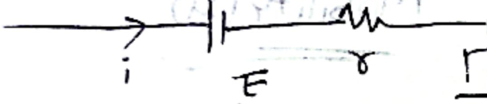
$R = \frac{\rho l}{A} = \frac{\rho l}{\pi R^2}$

2.

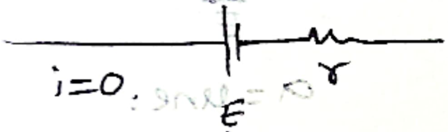
$R = \frac{\rho l}{\pi ab}$



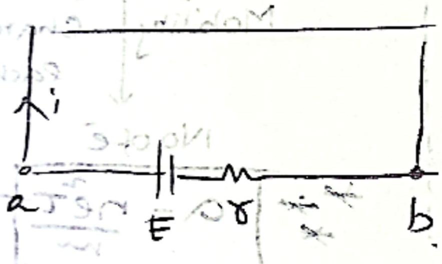
$\boxed{V = (E) - (ir)}$ — (i)



$\boxed{V = E + ir}$ — (ii)



$\boxed{V = E}$ — (iii)



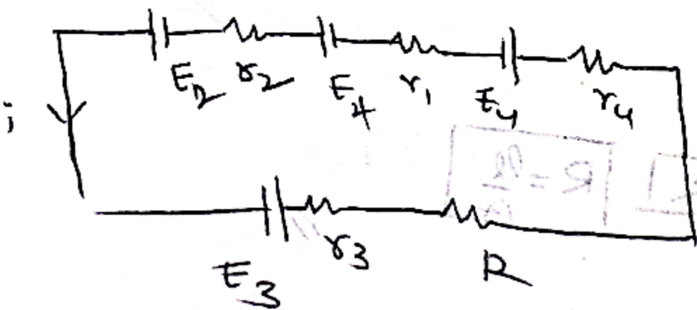
$V = 0$

let $V = V_a - V_b$ = Terminal Potential Droop...

Grouping of cells:

i. Parallel

ii. Series



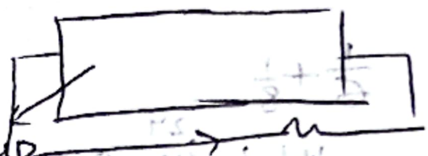
$i = \frac{E_1 + E_2 + E_3 - E_4}{R + r_1 + r_2 + r_3 + r_4}$

$10 = 17$.

Q. n identical cells:

$E = 10V$, $r = 2\Omega$. Out of 10 cells, 3 cells are connected by reverse polarity...

$70 - 30 \quad \frac{40}{20} \quad 2$.



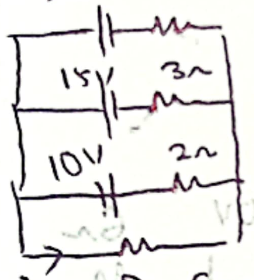
$V = iR$

$i = \frac{V}{R} = \frac{2}{10} = \frac{1}{5}$.

$i = 4/3 A$!!

Parallel Grouping of cells:

1.



find $i = ?$

$\frac{1}{4}+\frac{1}{3}+\frac{1}{2}$

$\frac{3}{4}+\frac{1}{3}$

$\frac{13}{12}$ $\frac{12}{13}$

$i = \frac{\Sigma \frac{E}{r}}{1+R\Sigma\frac{1}{r}} = \frac{\frac{20}{4}+\frac{15}{3}-\frac{10}{2}}{1+6\left(\frac{1}{4}+\frac{1}{3}+\frac{1}{12}\right)}$

$\frac{5}{\frac{15}{2}}$ $\frac{2}{3}$!!

$20 - 4i = \frac{2}{3}$

$= 4$

$15 - 3i = 4$

$3i = 11$

$-10 + 2i = -4$

$2i = 6$

$i = 3$

Power related Questions

$\boxed{dw = F dx}$ — (1)

$dw = P.A dx$

$dw = P.dV$

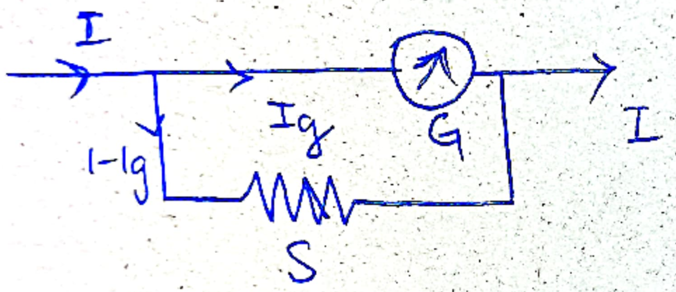
$$\frac{dw}{dV} = P$$

$$W = \int_{t_1}^{t_2} F dx$$

$\boxed{dx = V dt}$

$$a = V\frac{dx}{dx}$$

Derive by Equations and Get the relations...



$$I_g G = (I - I_g) S$$

$$\boxed{S = \frac{I_g G}{(I - I_g)}} \text{ — (1)}$$

Where

i. S → Shunt resistance.

ii. G → Galvanometer Coil in Ohm Ω

iii. I → Mesure Current in Amp(A)

iv. I_g → Mostly in mA